

Name : _____

HURLSTONE AGRICULTURAL HIGH SCHOOL

YEAR 12 2011 MATHEMATICS

ASSESSMENT TASK 1

Examiners: P. Biczo, S. Gee

General Instructions

- Reading time : 3 minutes
- **Working time : 40 minutes**
- Attempt **all** questions
- **Start a new sheet of paper for each question**
- All necessary working should be shown
- This paper contains 4 questions worth 8 marks each. Total Marks: **32 marks**
- Marks may not be awarded for careless or badly arranged work
- Board approved calculators may be used
- This examination paper must **not** be removed from the examination room

Question 1 (Start a new sheet of paper)

Marks

(a) Evaluate

(i) $\lim_{x \rightarrow \infty} \frac{x^2 - 3x - 1}{2x^2 - 1}$ **1**

(ii) $\lim_{x \rightarrow 0} \frac{x}{x^2 - 1}$ **1**

(iii) $\lim_{x \rightarrow 1} \frac{x - 1}{x^2 - 1}$ **1**

(b) Use the definition of the derivative by first principles,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

to show that if $f(x) = x^2 - 5x$, then $f'(x) = 2x - 5$ **3**

(iii) Draw a sketch of a possible curve that is differentiable in the domain $-a \leq x \leq a$. **2**

Question 2 (Start a new sheet of paper)**Marks**(a) Differentiate, with respect to x :

(i) $x^4 - 3x^2 + 5$ **1**

(ii) $(2x^2 + 5)^4$ **1**

(iii) $\frac{1}{\sqrt{x}}$ **1**

(iv) $(x^2 + 1)(3x - 1)^2$ **using the product rule** **2**

(v) $\frac{3x + 5}{4 - 5x}$ **2**

(b) Write down a version of the chain rule for finding the derivative of a function. **1****Question 3 (Start a new sheet of paper)**(a) Consider the curve $f(x) = x^3 - 6x^2 + 9x + 1$.(i) Find the coordinates of any stationary points and determine their nature. **3**(ii) Sketch the curve, showing this information. **1**(iii) Find any points of inflexion. **2**(b) By using calculus, show that the curve $y = \frac{3}{x}$ is decreasing for all values of x for which it is defined. Justify your answer. **2**

Question 4 (Start a new sheet of paper)**Marks**

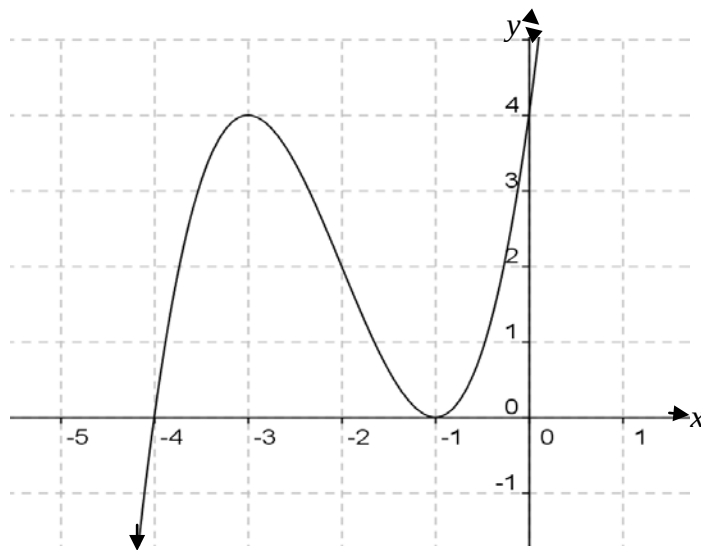
(a) Consider the curve $y = x^2 - 4x + 1$.

(i) Show that the gradient of the tangent at $x = 4$ is 4. **1**

(ii) Find the equation of the normal to the curve at the point where $x = 4$. **2**

(iii) At what point on the curve is the angle of inclination of the tangent 45° ? **2**

(b) The graph shown is $y = f(x)$. It is given that the only value for x at which $f''(x) = 0$ is when $x = -2$.



(i) For what values of x is $y = f(x)$ increasing? **1**

(ii) For what values of x is $f''(x) < 0$? **1**

(iii) Is the point $(-2, 2)$ a horizontal point of inflexion? **Justify** your answer. **1**

Outcomes Addressed in this Question

- P6 relates the derivative of a function to the slope of its graph.
P7 determines the derivative of a function through routine application of the rules of differentiation
P8 understands and uses the language and notation of calculus.

Outcome	Solutions	Marking Guidelines
	Question 1 a i) $\lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^2} - \frac{3x}{x^2} - \frac{2}{x^2}}{\frac{2x^2}{x^2} - \frac{1}{x^2}}$ $= \lim_{x \rightarrow \infty} \frac{1 - \frac{3}{x} - \frac{2}{x^2}}{2 - \frac{1}{x^2}}$ $= \frac{1}{2}$ ii) $\lim_{x \rightarrow 0} \frac{x}{x^2 - 1}$ $= \frac{0}{0 - 1} = 0$ iii) $\lim_{x \rightarrow 1} \frac{x - 1}{x^2 - 1}$ $= \lim_{x \rightarrow 1} \frac{x - 1}{(x - 1)(x + 1)}$ $= \lim_{x \rightarrow 1} \frac{1}{x + 1} = \frac{1}{2}$ b $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $f(x) = x^2 - 5x$ $f(x+h) = (x+h)^2 - 5(x+h) = x^2 + 2xh + h^2 - 5x - 5h$ $f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 5x - 5h - (x^2 - 5x)}{h}$ $= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 5h}{h}$ $= \lim_{h \rightarrow 0} (2x + h - 5)$ $= 2x - 5$ c	1 mark method leading to correct answer 1 mark method leading to correct answer 1 mark method leading to correct answer 3 marks correct method leading to correct conclusion 2 marks substantially correct solution 1 mark elementary progress towards correct solution
P8		
P6		
P5, 6	Any continuous smooth curve with end points at $x = -a$ and $x = a$.	1 mark suitable curve 1 mark clearly indicated end points.

Question 2

a

i) $\frac{d}{dx}(x^4 - 3x^2 + 5) = 4x^3 - 6x$

1 mark correct answer

ii) $\frac{d}{dx}[(2x^2 + 5)^4] = 4(2x^2 + 5)^3 \times 4x$
 $= 16(2x^2 + 5)^3$

1 mark correct answer

iii) $\frac{d}{dx}\left(\frac{1}{\sqrt{x}}\right) = \frac{d}{dx}(x^{-\frac{1}{2}})$
 $= -\frac{1}{2}x^{-\frac{3}{2}}$
 $= \frac{-1}{2\sqrt{x^3}}$

1 mark correct answer

iv) Let $y = (x^2 + 1)(3x - 1)^2$
 $u = x^2 + 1 \quad u' = 2x$
 $v = (3x - 1)^2 \quad v' = 6(3x - 1)$
 $\frac{d}{dx}(uv) = vu' + uv'$
 $= (3x - 1)^2 \times 2x + (x^2 + 1) \times 6(3x - 1)$
 $= 2(3x - 1)(x(3x - 1) + 3(x^2 + 1))$
 $= 2(3x - 1)(6x^2 - x + 3)$

2 marks correct method leading to correct answer

1 mark substantially correct solution

v) $\frac{d}{dx}\left(\frac{3x + 5}{4 - 5x}\right)$
 $u = 3x + 5 \quad v = 4 - 5x$
 $u' = 3 \quad v' = -5$
 $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$
 $\frac{d}{dx}\left(\frac{3x + 5}{4 - 5x}\right) = \frac{3(4 - 5x) - (-5)(3x + 5)}{(4 - 5x)^2}$
 $= \frac{37}{(4 - 5x)^2}$

2 marks correct method leading to correct answer

1 mark substantially correct solution

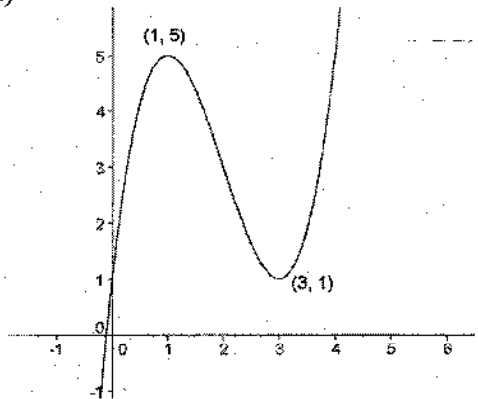
b

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

1 mark correct answer

P7

Outcomes Addressed in this Question**H6** Uses the derivative to determine features of the graph of a function**H5** applies appropriate techniques from the study of calculus and geometry to solve problems

Outcome	Solutions	Marking Guidelines								
H6	a)(i) $y = x^3 - 6x^2 + 9x + 1$ $y' = 3x^2 - 12x + 9 = 0$ for stationary points $\therefore 3(x-3)(x-1) = 0$ $\therefore$ stationary points at $x = 3$ and $x = 1$ $y'' = 6x - 12$ When $x = 1$, $y'' = -6 < 0 \therefore$ concave down & so relative maximum at $x = 1$ When $x = 3$, $y'' = 6 > 0 \therefore$ concave up & so relative minimum at $x = 3$ Maximum at (1,5), Minimum at (3,1) (ii) 	3 marks : puts $y' = 0$ and correctly solves; finds y values at stationary points; and correctly justifies and determines nature 2 marks: substantial progress towards correct solution 1 mark: significant progress towards correct solution								
H6		1 mark: correct graph or equivalent								
H6	(iii) Possible points of inflexion at $y'' = 0$ $\therefore 6x - 12 = 0$ $\therefore x = 2$ is a possible point of inflexion. <table border="1" data-bbox="378 1485 660 1565"> <tr> <td>x</td><td>1</td><td>2</td><td>3</td></tr> <tr> <td>y''</td><td>-6</td><td>0</td><td>+6</td></tr> </table> Since the concavity changes, $x = 2$ is a point of inflexion	x	1	2	3	y''	-6	0	+6	2 mark : finds pt.of inflexion & justifies concavity change 1 mark : one of above
x	1	2	3							
y''	-6	0	+6							
H5	(b) $y = \frac{3}{x} = 3x^{-1}$ $\frac{dy}{dx} = -3x^{-2} = \frac{-3}{x^2}$ Since x^2 is positive for all values of x (except $x = 0$ where it is undefined), $\frac{dy}{dx}$ is negative for all $x \neq 0$. $\therefore y$ is decreasing.	2 mark : correct derivative and justification 1 mark : one of above								

Year 11 Mathematics task 1 HSC 2010		
Question No. 4		Solutions and Marking Guidelines
Outcomes Addressed in this Question		
H5-applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems		
P6-relates the derivative of a function to the slope of its graph		
H6-uses the derivative to determine the features of the graph of a function		
P8-understands and uses the language and notation of calculus		
H7-uses the features of a graph to deduce information about the derivative		
Outcome	Solutions	Marking Guidelines
P6,P8	4. a)i) $y = x^2 - 4x + 1$ $\frac{dy}{dx} = 2x - 4$ at $x = 4$ $\frac{dy}{dx} = 2(4) - 4 = 4$ $\therefore$ gradient of tangent at $x = 4$ is 4.	1 mark for complete correct solution
H5	ii) Gradient of tangent is $m_1 = 4$ Gradient of normal is $m_1 = -\frac{1}{4}$ Equation of normal is: $y - 1 = -\frac{1}{4}(x - 4) \text{ or equivalent equation}$	2 marks for complete correct solution 1 mark for partial correct solution
H5,P8	iii) Gradient of tangent is $m_1 = \tan 45^\circ = 1$ $\frac{dy}{dx} = 2x - 4$ hence $2x - 4 = 1$ $\therefore x = \frac{5}{2}, \quad y = \left(\frac{5}{2}\right)^2 - 4\left(\frac{5}{2}\right) + 1 = -\frac{11}{4}$ $\therefore$ The tangent is inclined at an angle of 45° to the curve at the point $\left(\frac{5}{2}, -\frac{11}{4}\right)$.	2 marks for complete correct solution 1 mark for partial correct solution [Students must name both coordinates of the point]
H6,H7	b)i) $x < -3, x > -1$	1 mark for complete correct solution
H6, H7	ii) $x < -2$	1 mark for complete correct solution
H6, H7	iii) No, since it is not a stationary point i.e. $\frac{dy}{dx} \neq 0$ at $(-2, 2)$.	1 mark for complete correct solution